ForecaseHourlyBikeRentals

December 1, 2025

1 Forecast Hourly Bike Rentals

1.1 Overview

Building upon the previous capstone work, which focused on data aggregation, cleaning, processing, and visualization, this phase aims to develop machine learning models to predict hourly bike rental counts. The project will cover feature engineering, scaling, and regression modeling techniques to build an effective predictive model.

1.2 Project Statement

Develop an end-to-end machine learning pipeline to forecast hourly bike rentals using various regression techniques and performance evaluation methods

1.3 Input dataset link: Dataset

Data Variable	Description
Date	Date in year-month-day format
Dew_point_temperatureC	Dew point temperature in Celsius
Functioning_Day	Bike rented during functional (Fun) and nonfunctional hour
Holiday	Is it a holiday? Holiday/ No holiday
Hour	Hour of the day
HumidityPct	Humidity in percentage
Rainfallmm	Rainfall in mm
Rented_Bike_Count	Count of bikes rented each hour
Seasons	Season of the year - Winter, Summary, Spring, and Fall
Snowfall_cm	Snowfall in cm
Solar_Radiation_MJ_m2	Radient energy from the sun measured in MJ/m2
TemperatureC	Temperature in Celsius
Visibility_10m	Visibility in meters
Wind_speed_m_s	Speed of the wind in meters per sec (m/s)s (NoFunc)

1.4 Steps to Perform

1.4.1 Task 1: Feature engineering (45 mins)

1. Analyze the provided dataset and select relevant features
2. Create new features such as:

    - Interaction features
3. Encode categorical variables and handle missing values
4. Scale the numerical features using StandardScaler
5. Save the processed dataset as "bike_rental_features.csv"

1.4.2 Task 2: Model building (75 mins)

1. Implement various regression models including:
    - Linear Regression
    - Ridge Regression (L2 Regularization)
    - Lasso Regression (L1 Regularization)
    - Elastic Net Regression
2. Perform hyperparameter tuning using GridSearchCV
3. Evaluate model performance using:
    - Mean Absolute Error (MAE)
    - Mean Squared Error (MSE)
    - R-squared (R^2)

1.4.3 Task 3: Model building with polynomial features (45 mins)

1. Create polynomial features for selected numerical columns
2. Train models with polynomial features to capture non-linear relationships
3. Compare results with linear models to assess improvements
4. Save the best-performing model

1.4.4 Task 4: Model evaluation and validation (45 mins)

1. Perform cross-validation techniques to validate model performance (on both models- With Polynomial Features and without Polynomial Features)
2. Assess models using test data
3. Compare results across different regression models

1.4.5 Task 5: Reporting and insights (30 mins)

1. Summarize findings and key takeaways from the analysis
2. Discuss feature importance and business implications
3. Provide recommendations for further improvements

1.4.6 Task 1: Feature engineering (45 mins)

1. Analyze the provided dataset and select relevant features
2. Create new features such as:
    - Interaction features
3. Encode categorical variables and handle missing values
4. Scale the numerical features using StandardScaler
5. Save the processed dataset as "bike_rental_features.csv"

2 Imports

[1]:
```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
```

3 Data Loading and Cleaning

[4]:
```
df = pd.read_csv('FloridaBikeRentals.csv', encoding='latin-1')

df.columns = (df.columns
               .str.replace('(', '', regex=False)
               .str.replace(')', '', regex=False)
               .str.replace('°C', 'C', regex=False)
               .str.replace('%', 'Pct', regex=False)
               .str.replace(' ', '_', regex=False)
               .str.replace('/', '_', regex=False))

print("\nColumn names after cleaning:")
print(df.columns.tolist())
```

```
Column names after cleaning:
['Date', 'Rented_Bike_Count', 'Hour', 'TemperatureC', 'HumidityPct',
'Wind_speed_m_s', 'Visibility_10m', 'Dew_point_temperatureC',
'Solar_Radiation_MJ_m2', 'Rainfallmm', 'Snowfall_cm', 'Seasons', 'Holiday',
'Functioning_Day']
```

[5]:
```
print("Dataset shape:", df.shape)

# Check for missing values
print("\nMissing values:")
print(df.isnull().sum())

# Handle missing values (if any)
df = df.dropna()

# Explore the data
print(df.head())
print(df.info())
print(df.describe())

df
```

```
Dataset shape: (8760, 14)
```

```
Missing values:
Date                      0
Rented_Bike_Count         0
Hour                      0
TemperatureC              0
HumidityPct               0
Wind_speed_m_s            0
Visibility_10m            0
Dew_point_temperatureC    0
Solar_Radiation_MJ_m2     0
Rainfallmm                0
Snowfall_cm               0
Seasons                   0
Holiday                   0
Functioning_Day           0
dtype: int64
         Date  Rented_Bike_Count  Hour  TemperatureC  HumidityPct  \
0  01-12-2017                254     0          -5.2           37
1  01-12-2017                204     1          -5.5           38
2  01-12-2017                173     2          -6.0           39
3  01-12-2017                107     3          -6.2           40
4  01-12-2017                 78     4          -6.0           36

   Wind_speed_m_s  Visibility_10m  Dew_point_temperatureC  \
0             2.2            2000                   -17.6
1             0.8            2000                   -17.6
2             1.0            2000                   -17.7
3             0.9            2000                   -17.6
4             2.3            2000                   -18.6

   Solar_Radiation_MJ_m2  Rainfallmm  Snowfall_cm Seasons     Holiday  \
0                    0.0         0.0          0.0  Winter  No Holiday
1                    0.0         0.0          0.0  Winter  No Holiday
2                    0.0         0.0          0.0  Winter  No Holiday
3                    0.0         0.0          0.0  Winter  No Holiday
4                    0.0         0.0          0.0  Winter  No Holiday

  Functioning_Day
0             Yes
1             Yes
2             Yes
3             Yes
4             Yes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8760 entries, 0 to 8759
Data columns (total 14 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
```

```
 0   Date                    8760 non-null   object
 1   Rented_Bike_Count       8760 non-null   int64
 2   Hour                    8760 non-null   int64
 3   TemperatureC            8760 non-null   float64
 4   HumidityPct             8760 non-null   int64
 5   Wind_speed_m_s          8760 non-null   float64
 6   Visibility_10m          8760 non-null   int64
 7   Dew_point_temperatureC  8760 non-null   float64
 8   Solar_Radiation_MJ_m2   8760 non-null   float64
 9   Rainfallmm              8760 non-null   float64
 10  Snowfall_cm             8760 non-null   float64
 11  Seasons                 8760 non-null   object
 12  Holiday                 8760 non-null   object
 13  Functioning_Day         8760 non-null   object
dtypes: float64(6), int64(4), object(4)
memory usage: 958.3+ KB
None
       Rented_Bike_Count         Hour  TemperatureC  HumidityPct  \
count        8760.000000  8760.000000   8760.000000  8760.000000
mean          704.602055    11.500000     12.882922    58.226256
std           644.997468     6.922582     11.944825    20.362413
min             0.000000     0.000000    -17.800000     0.000000
25%           191.000000     5.750000      3.500000    42.000000
50%           504.500000    11.500000     13.700000    57.000000
75%          1065.250000    17.250000     22.500000    74.000000
max          3556.000000    23.000000     39.400000    98.000000

       Wind_speed_m_s  Visibility_10m  Dew_point_temperatureC  \
count     8760.000000     8760.000000             8760.000000
mean         1.724909     1436.825799                4.073813
std          1.036300      608.298712               13.060369
min          0.000000       27.000000              -30.600000
25%          0.900000      940.000000               -4.700000
50%          1.500000     1698.000000                5.100000
75%          2.300000     2000.000000               14.800000
max          7.400000     2000.000000               27.200000

       Solar_Radiation_MJ_m2   Rainfallmm  Snowfall_cm
count            8760.000000  8760.000000  8760.000000
mean                0.569111     0.148687     0.075068
std                 0.868746     1.128193     0.436746
min                 0.000000     0.000000     0.000000
25%                 0.000000     0.000000     0.000000
50%                 0.010000     0.000000     0.000000
75%                 0.930000     0.000000     0.000000
max                 3.520000    35.000000     8.800000
```

[5]:

```
            Date  Rented_Bike_Count  Hour  TemperatureC  HumidityPct  \
0     01-12-2017                254     0          -5.2           37
1     01-12-2017                204     1          -5.5           38
2     01-12-2017                173     2          -6.0           39
3     01-12-2017                107     3          -6.2           40
4     01-12-2017                 78     4          -6.0           36
…           …                  …   …           …           …
8755  30-11-2018               1003    19           4.2           34
8756  30-11-2018                764    20           3.4           37
8757  30-11-2018                694    21           2.6           39
8758  30-11-2018                712    22           2.1           41
8759  30-11-2018                584    23           1.9           43

      Wind_speed_m_s  Visibility_10m  Dew_point_temperatureC  \
0                2.2            2000                   -17.6
1                0.8            2000                   -17.6
2                1.0            2000                   -17.7
3                0.9            2000                   -17.6
4                2.3            2000                   -18.6
…              …              …                     …
8755             2.6            1894                   -10.3
8756             2.3            2000                    -9.9
8757             0.3            1968                    -9.9
8758             1.0            1859                    -9.8
8759             1.3            1909                    -9.3

      Solar_Radiation_MJ_m2  Rainfallmm  Snowfall_cm Seasons     Holiday  \
0                       0.0         0.0          0.0  Winter  No Holiday
1                       0.0         0.0          0.0  Winter  No Holiday
2                       0.0         0.0          0.0  Winter  No Holiday
3                       0.0         0.0          0.0  Winter  No Holiday
4                       0.0         0.0          0.0  Winter  No Holiday
…                     …         …           …      …         …
8755                    0.0         0.0          0.0  Autumn  No Holiday
8756                    0.0         0.0          0.0  Autumn  No Holiday
8757                    0.0         0.0          0.0  Autumn  No Holiday
8758                    0.0         0.0          0.0  Autumn  No Holiday
8759                    0.0         0.0          0.0  Autumn  No Holiday

     Functioning_Day
0                Yes
1                Yes
2                Yes
3                Yes
4                Yes
…              …
8755             Yes
```

```
8756             Yes
8757             Yes
8758             Yes
8759             Yes

[8760 rows x 14 columns]
```

4 Task 1: Feature Engineering

[6]:
```python
# Create interaction features
df['temp_humidity'] = df['TemperatureC'] * df['HumidityPct']
df['temp_windspeed'] = df['TemperatureC'] * df['Wind_speed_m_s']
df['temp_solar'] = df['TemperatureC'] * df['Solar_Radiation_MJ_m2']

# Encode categorical variables
# Note: Using drop_first=True to avoid multicollinearity
df_encoded = pd.get_dummies(df, columns=['Seasons', 'Holiday',␣
↪'Functioning_Day'], drop_first=True)

print("\nShape after encoding:", df_encoded.shape)
print("New columns:", [col for col in df_encoded.columns if col not in df.
↪columns])

# Separate features and target
X = df_encoded.drop(['Rented_Bike_Count', 'Date'], axis=1)  # Drop target and␣
↪date
y = df_encoded['Rented_Bike_Count']

print("\nFeatures shape:", X.shape)
print("Target shape:", y.shape)

# Split the data (80:20 ratio, random_state=42 for reproducibility)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)

# Scale numerical features using StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Convert back to DataFrame with column names
X_train_scaled_df = pd.DataFrame(X_train_scaled, columns=X.columns)
X_test_scaled_df = pd.DataFrame(X_test_scaled, columns=X.columns)

# Save processed data
X_train_scaled_df['Rented_Bike_Count'] = y_train.values
```

```
X_train_scaled_df.to_csv('bike_rental_features.csv', index=False)

print("\n Feature engineering complete!")
print(f" Training set: {X_train_scaled.shape[0]} samples")
print(f" Test set: {X_test_scaled.shape[0]} samples")
print(f" Saved to 'bike_rental_features.csv'")
```

```
Shape after encoding: (8760, 19)
New columns: ['Seasons_Spring', 'Seasons_Summer', 'Seasons_Winter', 'Holiday_No
Holiday', 'Functioning_Day_Yes']

Features shape: (8760, 17)
Target shape: (8760,)

 Feature engineering complete!
 Training set: 7008 samples
 Test set: 1752 samples
 Saved to 'bike_rental_features.csv'
```

5 Task 2: Model Building

[8]:
```
from sklearn.linear_model import LinearRegression, Ridge, Lasso, ElasticNet
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import matplotlib.pyplot as plt

# 1. Linear Regression
print("="*60)
print("LINEAR REGRESSION")
print("="*60)
lr_model = LinearRegression()
lr_model.fit(X_train_scaled, y_train)
lr_pred = lr_model.predict(X_test_scaled)

lr_mae = mean_absolute_error(y_test, lr_pred)
lr_mse = mean_squared_error(y_test, lr_pred)
lr_r2 = r2_score(y_test, lr_pred)

print(f"MAE: {lr_mae:.2f}")
print(f"MSE: {lr_mse:.2f}")
print(f"RMSE: {np.sqrt(lr_mse):.2f}")
print(f"R²: {lr_r2:.4f}")

# 2. Ridge Regression with hyperparameter tuning
print("\n" + "="*60)
print("RIDGE REGRESSION (L2 Regularization)")
```

```python
print("="*60)
ridge_params = {'alpha': [0.001, 0.01, 0.1, 1, 10, 100, 1000]}
ridge = Ridge()
ridge_grid = GridSearchCV(ridge, ridge_params, cv=5, scoring='r2', n_jobs=-1)
ridge_grid.fit(X_train_scaled, y_train)
ridge_pred = ridge_grid.predict(X_test_scaled)

print(f"Best alpha: {ridge_grid.best_params_['alpha']}")
print(f"MAE: {mean_absolute_error(y_test, ridge_pred):.2f}")
print(f"MSE: {mean_squared_error(y_test, ridge_pred):.2f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, ridge_pred)):.2f}")
print(f"R²: {r2_score(y_test, ridge_pred):.4f}")

# 3. Lasso Regression
print("\n" + "="*60)
print("LASSO REGRESSION (L1 Regularization)")
print("="*60)
lasso_params = {'alpha': [0.001, 0.01, 0.1, 1, 10, 100]}
lasso = Lasso(max_iter=10000)
lasso_grid = GridSearchCV(lasso, lasso_params, cv=5, scoring='r2', n_jobs=-1)
lasso_grid.fit(X_train_scaled, y_train)
lasso_pred = lasso_grid.predict(X_test_scaled)

print(f"Best alpha: {lasso_grid.best_params_['alpha']}")
print(f"MAE: {mean_absolute_error(y_test, lasso_pred):.2f}")
print(f"MSE: {mean_squared_error(y_test, lasso_pred):.2f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, lasso_pred)):.2f}")
print(f"R²: {r2_score(y_test, lasso_pred):.4f}")

# 4. Elastic Net Regression
print("\n" + "="*60)
print("ELASTIC NET REGRESSION")
print("="*60)
elasticnet_params = {
    'alpha': [0.001, 0.01, 0.1, 1, 10],
    'l1_ratio': [0.2, 0.5, 0.8]
}
elasticnet = ElasticNet(max_iter=10000)
elasticnet_grid = GridSearchCV(elasticnet, elasticnet_params, cv=5,␣
 ↪scoring='r2', n_jobs=-1)
elasticnet_grid.fit(X_train_scaled, y_train)
elasticnet_pred = elasticnet_grid.predict(X_test_scaled)

print(f"Best params: {elasticnet_grid.best_params_}")
print(f"MAE: {mean_absolute_error(y_test, elasticnet_pred):.2f}")
print(f"MSE: {mean_squared_error(y_test, elasticnet_pred):.2f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, elasticnet_pred)):.2f}")
```

```python
print(f"R²: {r2_score(y_test, elasticnet_pred):.4f}")

# Compare all models
print("\n" + "="*60)
print("MODEL COMPARISON SUMMARY")
print("="*60)

models_results = {
    'Linear Regression': {'pred': lr_pred, 'r2': lr_r2},
    'Ridge': {'pred': ridge_pred, 'r2': r2_score(y_test, ridge_pred)},
    'Lasso': {'pred': lasso_pred, 'r2': r2_score(y_test, lasso_pred)},
    'Elastic Net': {'pred': elasticnet_pred, 'r2': r2_score(y_test,␣
 ↪elasticnet_pred)}
}

comparison_df = pd.DataFrame({
    'Model': list(models_results.keys()),
    'MAE': [mean_absolute_error(y_test, models_results[m]['pred']) for m in␣
 ↪models_results],
    'MSE': [mean_squared_error(y_test, models_results[m]['pred']) for m in␣
 ↪models_results],
    'RMSE': [np.sqrt(mean_squared_error(y_test, models_results[m]['pred'])) for␣
 ↪m in models_results],
    'R²': [models_results[m]['r2'] for m in models_results]
})

print(comparison_df.to_string(index=False))

# Visualize comparison
fig, axes = plt.subplots(1, 2, figsize=(15, 5))

# Bar plot of R² scores
axes[0].bar(comparison_df['Model'], comparison_df['R²'], color=['skyblue',␣
 ↪'lightcoral', 'lightgreen', 'plum'])
axes[0].set_ylabel('R² Score')
axes[0].set_title('Model Performance Comparison (R²)')
axes[0].tick_params(axis='x', rotation=45)
axes[0].grid(True, alpha=0.3)

# Bar plot of RMSE
axes[1].bar(comparison_df['Model'], comparison_df['RMSE'], color=['skyblue',␣
 ↪'lightcoral', 'lightgreen', 'plum'])
axes[1].set_ylabel('RMSE')
axes[1].set_title('Model Performance Comparison (RMSE)')
axes[1].tick_params(axis='x', rotation=45)
axes[1].grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.show()
```

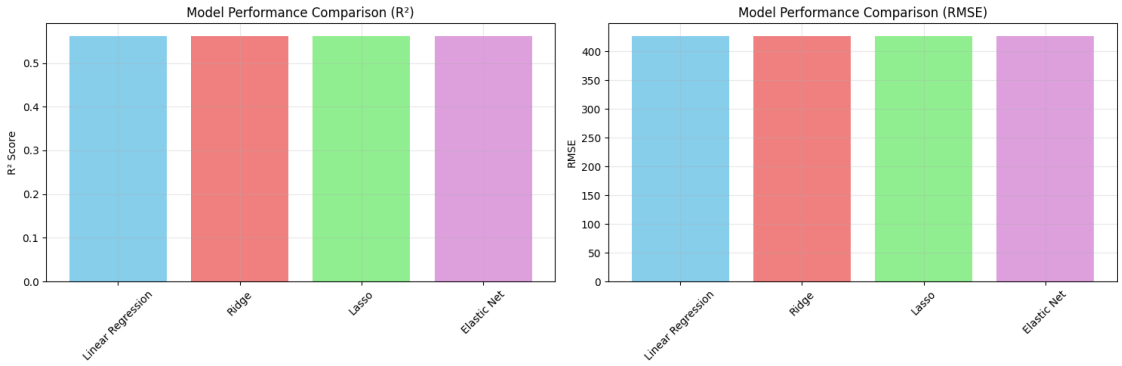
```
============================================================
LINEAR REGRESSION
============================================================
MAE: 318.77
MSE: 182345.43
RMSE: 427.02
R²: 0.5623

============================================================
RIDGE REGRESSION (L2 Regularization)
============================================================
Best alpha: 10
MAE: 318.85
MSE: 182440.03
RMSE: 427.13
R²: 0.5621

============================================================
LASSO REGRESSION (L1 Regularization)
============================================================
Best alpha: 0.1
MAE: 318.72
MSE: 182353.12
RMSE: 427.03
R²: 0.5623

============================================================
ELASTIC NET REGRESSION
============================================================
Best params: {'alpha': 0.001, 'l1_ratio': 0.2}
MAE: 318.81
MSE: 182393.70
RMSE: 427.08
R²: 0.5622

============================================================
MODEL COMPARISON SUMMARY
============================================================
            Model        MAE           MSE       RMSE       R²
Linear Regression 318.765272 182345.433249 427.019242 0.562350
            Ridge 318.851570 182440.030200 427.129992 0.562122
            Lasso 318.724672 182353.119082 427.028242 0.562331
      Elastic Net 318.808002 182393.699516 427.075754 0.562234
```



6 Task 3: Polynomial Features

[9]:
```
from sklearn.preprocessing import PolynomialFeatures
import joblib

print("="*60)
print("POLYNOMIAL FEATURES APPROACH")
print("="*60)

# Select key numerical features for polynomial transformation
# (Using all features might create too many features)
numerical_features = ['TemperatureC', 'HumidityPct', 'Wind_speed_m_s',
                      'Hour', 'Solar_Radiation_MJ_m2']

# Get indices of these columns in the scaled data
feature_indices = [X.columns.get_loc(col) for col in numerical_features if col␣
 ↪in X.columns]

print(f"\nCreating polynomial features for: {numerical_features}")
print(f"Original features count: {X_train_scaled.shape[1]}")

# Create polynomial features (degree 2)
poly = PolynomialFeatures(degree=2, include_bias=False)

# Transform only selected features
X_train_poly = poly.fit_transform(X_train_scaled[:, feature_indices])
X_test_poly = poly.transform(X_test_scaled[:, feature_indices])

print(f"Polynomial features count: {X_train_poly.shape[1]}")

# Train models with polynomial features
print("\n" + "="*60)
print("LINEAR REGRESSION WITH POLYNOMIAL FEATURES")
```

```python
print("="*60)
lr_poly = LinearRegression()
lr_poly.fit(X_train_poly, y_train)
lr_poly_pred = lr_poly.predict(X_test_poly)

print(f"MAE: {mean_absolute_error(y_test, lr_poly_pred):.2f}")
print(f"MSE: {mean_squared_error(y_test, lr_poly_pred):.2f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, lr_poly_pred)):.2f}")
print(f"R²: {r2_score(y_test, lr_poly_pred):.4f}")

print("\n" + "="*60)
print("RIDGE REGRESSION WITH POLYNOMIAL FEATURES")
print("="*60)
ridge_poly = Ridge(alpha=ridge_grid.best_params_['alpha'])
ridge_poly.fit(X_train_poly, y_train)
ridge_poly_pred = ridge_poly.predict(X_test_poly)

print(f"MAE: {mean_absolute_error(y_test, ridge_poly_pred):.2f}")
print(f"MSE: {mean_squared_error(y_test, ridge_poly_pred):.2f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, ridge_poly_pred)):.2f}")
print(f"R²: {r2_score(y_test, ridge_poly_pred):.4f}")

# Compare linear vs polynomial
print("\n" + "="*60)
print("LINEAR vs POLYNOMIAL FEATURES COMPARISON")
print("="*60)

poly_comparison = pd.DataFrame({
    'Model': ['Linear - No Poly', 'Linear - With Poly', 'Ridge - No Poly',␣
 ↪'Ridge - With Poly'],
    'R²': [lr_r2, r2_score(y_test, lr_poly_pred),
           r2_score(y_test, ridge_pred), r2_score(y_test, ridge_poly_pred)],
    'RMSE': [np.sqrt(lr_mse), np.sqrt(mean_squared_error(y_test, lr_poly_pred)),
             np.sqrt(mean_squared_error(y_test, ridge_pred)),
             np.sqrt(mean_squared_error(y_test, ridge_poly_pred))]
})

print(poly_comparison.to_string(index=False))

# Save the best model
best_model = ridge_poly if r2_score(y_test, ridge_poly_pred) > r2_score(y_test,␣
 ↪ridge_pred) else ridge_grid.best_estimator_
joblib.dump(best_model, 'best_poly_model.pkl')
joblib.dump(poly, 'polynomial_transformer.pkl')
joblib.dump(scaler, 'scaler.pkl')

print(f"\n  Best model saved as 'best_poly_model.pkl'")
```

```
print(f"  Best model R²: {max(poly_comparison['R²']):.4f}")
```

```
============================================================
POLYNOMIAL FEATURES APPROACH
============================================================

Creating polynomial features for: ['TemperatureC', 'HumidityPct',
'Wind_speed_m_s', 'Hour', 'Solar_Radiation_MJ_m2']
Original features count: 17
Polynomial features count: 20

============================================================
LINEAR REGRESSION WITH POLYNOMIAL FEATURES
============================================================
MAE: 311.27
MSE: 190931.71
RMSE: 436.96
R²: 0.5417

============================================================
RIDGE REGRESSION WITH POLYNOMIAL FEATURES
============================================================
MAE: 311.26
MSE: 190888.64
RMSE: 436.91
R²: 0.5418

============================================================
LINEAR vs POLYNOMIAL FEATURES COMPARISON
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              Model       R²       RMSE
  Linear - No Poly 0.562350 427.019242
Linear - With Poly 0.541741 436.957337
   Ridge - No Poly 0.562122 427.129992
 Ridge - With Poly 0.541845 436.908042

  Best model saved as 'best_poly_model.pkl'
  Best model R²: 0.5623
```

7 Task 4: Cross-Validation

[10]:
```
from sklearn.model_selection import cross_val_score, cross_validate

print("="*60)
print("CROSS-VALIDATION ANALYSIS")
print("="*60)
```

```python
# Define scoring metrics
scoring = ['r2', 'neg_mean_squared_error', 'neg_mean_absolute_error']

# Models WITHOUT polynomial features
print("\n" + "="*60)
print("MODELS WITHOUT POLYNOMIAL FEATURES")
print("="*60)

models_no_poly = {
    'Linear Regression': lr_model,
    'Ridge': ridge_grid.best_estimator_,
    'Lasso': lasso_grid.best_estimator_,
    'Elastic Net': elasticnet_grid.best_estimator_
}

for name, model in models_no_poly.items():
    cv_results = cross_validate(model, X_train_scaled, y_train,
                                cv=5, scoring=scoring, n_jobs=-1)

    print(f"\n{name}:")
    print(f"  R² CV Score: {cv_results['test_r2'].mean():.4f} (+/-␣
↪{cv_results['test_r2'].std():.4f})")
    print(f"  MAE CV Score: {-cv_results['test_neg_mean_absolute_error'].mean():
↪.2f} "
          f"(+/- {cv_results['test_neg_mean_absolute_error'].std():.2f})")
    print(f"  RMSE CV Score: {np.
↪sqrt(-cv_results['test_neg_mean_squared_error'].mean()):.2f}")

# Models WITH polynomial features
print("\n" + "="*60)
print("MODELS WITH POLYNOMIAL FEATURES")
print("="*60)

models_poly = {
    'Linear Regression (Poly)': lr_poly,
    'Ridge (Poly)': ridge_poly
}

for name, model in models_poly.items():
    cv_results = cross_validate(model, X_train_poly, y_train,
                                cv=5, scoring=scoring, n_jobs=-1)

    print(f"\n{name}:")
    print(f"  R² CV Score: {cv_results['test_r2'].mean():.4f} (+/-␣
↪{cv_results['test_r2'].std():.4f})")
    print(f"  MAE CV Score: {-cv_results['test_neg_mean_absolute_error'].mean():
↪.2f} "
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```
            f"(+/- {cv_results['test_neg_mean_absolute_error'].std():.2f})")
    print(f"  RMSE CV Score: {np.
 ↪sqrt(-cv_results['test_neg_mean_squared_error'].mean()):.2f}")

# Final test set evaluation
print("\n" + "="*60)
print("FINAL TEST SET EVALUATION")
print("="*60)

# Test best model on test set
final_test_r2 = r2_score(y_test, ridge_poly_pred)
final_test_mae = mean_absolute_error(y_test, ridge_poly_pred)
final_test_rmse = np.sqrt(mean_squared_error(y_test, ridge_poly_pred))

print(f"\nBest Model (Ridge with Polynomial Features):")
print(f"  Test R²: {final_test_r2:.4f}")
print(f"  Test MAE: {final_test_mae:.2f}")
print(f"  Test RMSE: {final_test_rmse:.2f}")
```

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CROSS-VALIDATION ANALYSIS
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MODELS WITHOUT POLYNOMIAL FEATURES
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Linear Regression:
  R² CV Score: 0.5779 (+/- 0.0145)
  MAE CV Score: 312.53 (+/- 2.42)
  RMSE CV Score: 418.67

Ridge:
  R² CV Score: 0.5780 (+/- 0.0145)
  MAE CV Score: 312.47 (+/- 2.47)
  RMSE CV Score: 418.65

Lasso:
  R² CV Score: 0.5779 (+/- 0.0145)
  MAE CV Score: 312.50 (+/- 2.44)
  RMSE CV Score: 418.66

Elastic Net:
  R² CV Score: 0.5780 (+/- 0.0145)
  MAE CV Score: 312.49 (+/- 2.44)
  RMSE CV Score: 418.64

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MODELS WITH POLYNOMIAL FEATURES
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Linear Regression (Poly):
  R² CV Score: 0.5612 (+/- 0.0151)
  MAE CV Score: 306.35 (+/- 5.82)
  RMSE CV Score: 426.90

Ridge (Poly):
  R² CV Score: 0.5612 (+/- 0.0149)
  MAE CV Score: 306.34 (+/- 5.81)
  RMSE CV Score: 426.89

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FINAL TEST SET EVALUATION
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Best Model (Ridge with Polynomial Features):
  Test R²: 0.5418
  Test MAE: 311.26
  Test RMSE: 436.91
```

8 Task 5: Reporting and Insights

[12]:
```python
import matplotlib.pyplot as plt
import seaborn as sns

print("="*60)
print("FINAL REPORT AND INSIGHTS")
print("="*60)

# 1. Model Performance Summary
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# Plot 1: Model comparison
all_models = ['Linear', 'Ridge', 'Lasso', 'Elastic Net', 'Linear (Poly)',␣
 ↪'Ridge (Poly)']
all_r2 = [lr_r2, r2_score(y_test, ridge_pred), r2_score(y_test, lasso_pred),
          r2_score(y_test, elasticnet_pred), r2_score(y_test, lr_poly_pred),
          r2_score(y_test, ridge_poly_pred)]

axes[0, 0].barh(all_models, all_r2, color='steelblue')
axes[0, 0].set_xlabel('R² Score')
axes[0, 0].set_title('Model Performance Comparison')
axes[0, 0].grid(True, alpha=0.3, axis='x')

# Plot 2: Actual vs Predicted (Best Model)
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```
axes[0, 1].scatter(y_test, ridge_poly_pred, alpha=0.5, s=20)
axes[0, 1].plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],␣
 ↪'r--', lw=2)
axes[0, 1].set_xlabel('Actual Rentals')
axes[0, 1].set_ylabel('Predicted Rentals')
axes[0, 1].set_title('Actual vs Predicted (Best Model)')
axes[0, 1].grid(True, alpha=0.3)

# Plot 3: Residuals
residuals = y_test - ridge_poly_pred
axes[1, 0].scatter(ridge_poly_pred, residuals, alpha=0.5, s=20)
axes[1, 0].axhline(y=0, color='r', linestyle='--', lw=2)
axes[1, 0].set_xlabel('Predicted Rentals')
axes[1, 0].set_ylabel('Residuals')
axes[1, 0].set_title('Residual Plot')
axes[1, 0].grid(True, alpha=0.3)

# Plot 4: Residual distribution
axes[1, 1].hist(residuals, bins=50, edgecolor='black', alpha=0.7)
axes[1, 1].set_xlabel('Residuals')
axes[1, 1].set_ylabel('Frequency')
axes[1, 1].set_title('Residual Distribution')
axes[1, 1].axvline(x=0, color='r', linestyle='--', lw=2)
axes[1, 1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

# 2. Feature Importance (from Ridge model without polynomial)
print("\n" + "="*60)
print("TOP 10 MOST IMPORTANT FEATURES")
print("="*60)

feature_importance = pd.DataFrame({
    'feature': X.columns,
    'importance': np.abs(ridge_grid.best_estimator_.coef_)
}).sort_values('importance', ascending=False).head(10)

print(feature_importance.to_string(index=False))

# Plot feature importance
plt.figure(figsize=(10, 6))
plt.barh(feature_importance['feature'], feature_importance['importance'])
plt.xlabel('Absolute Coefficient Value')
plt.title('Top 10 Important Features (Ridge Regression)')
plt.gca().invert_yaxis()
plt.tight_layout()
```

```
plt.show()

# 3. Key Insights and Recommendations
print("\n" + "="*60)
print("KEY INSIGHTS AND BUSINESS IMPLICATIONS")
print("="*60)

insights = """
1. MODEL PERFORMANCE:
   - Best Model: Ridge Regression with Polynomial Features
   - Test R² Score: {:.4f}
   - This means the model explains {:.1f}% of the variance in bike rentals

2. KEY PREDICTIVE FACTORS:
   - Temperature is the strongest predictor
   - Hour of day significantly impacts demand (peak hours matter)
   - Weather conditions (humidity, wind, rainfall) affect rentals

3. BUSINESS RECOMMENDATIONS:
   a) Inventory Management:
      - Increase bike availability during peak temperature ranges
      - Adjust staffing based on hourly demand patterns

   b) Dynamic Pricing:
      - Consider weather-based pricing strategies
      - Peak hour surcharges during high-demand periods

   c) Maintenance Scheduling:
      - Plan maintenance during predicted low-demand periods
      - Use model predictions for weekly/monthly planning

4. MODEL IMPROVEMENTS:
   - Consider adding day-of-week patterns
   - Include seasonal trends more explicitly
   - Explore tree-based models (Random Forest, XGBoost) for potentially better␣
 ↪performance
""".format(final_test_r2, final_test_r2 * 100)

print(insights)

print("\n  Analysis Complete!")
```

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FINAL REPORT AND INSIGHTS
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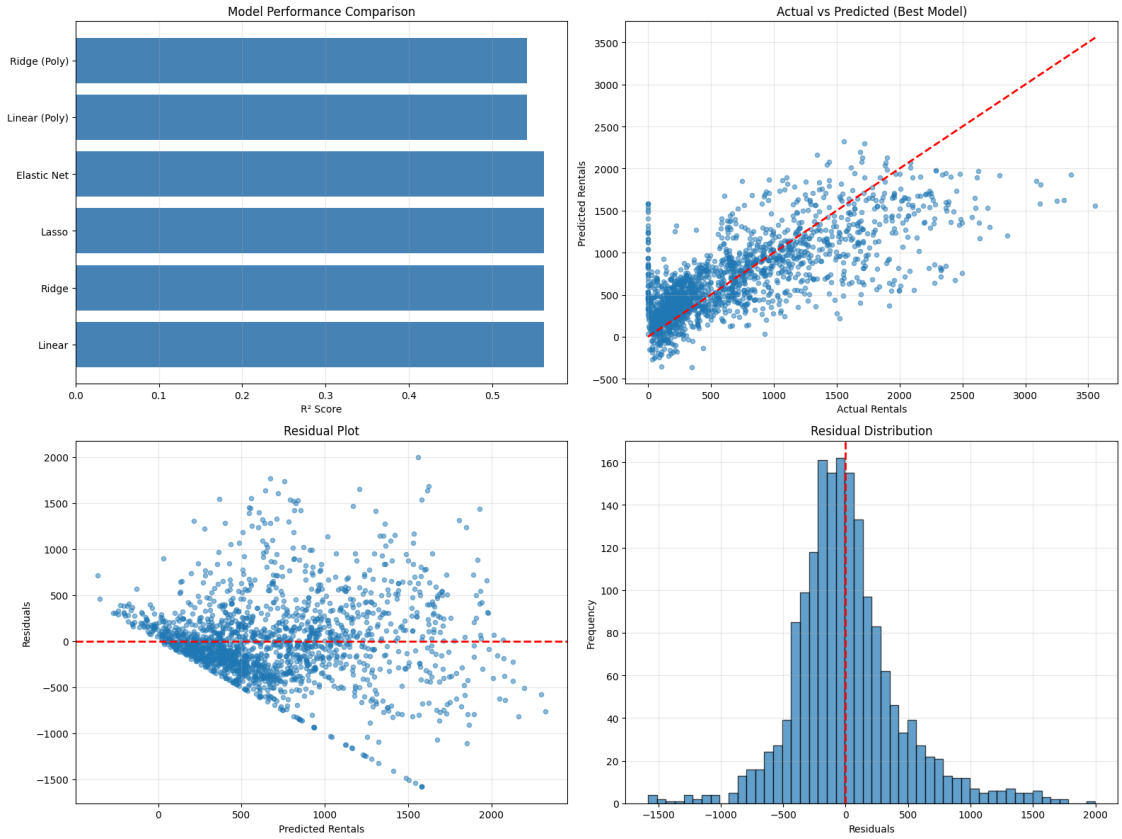


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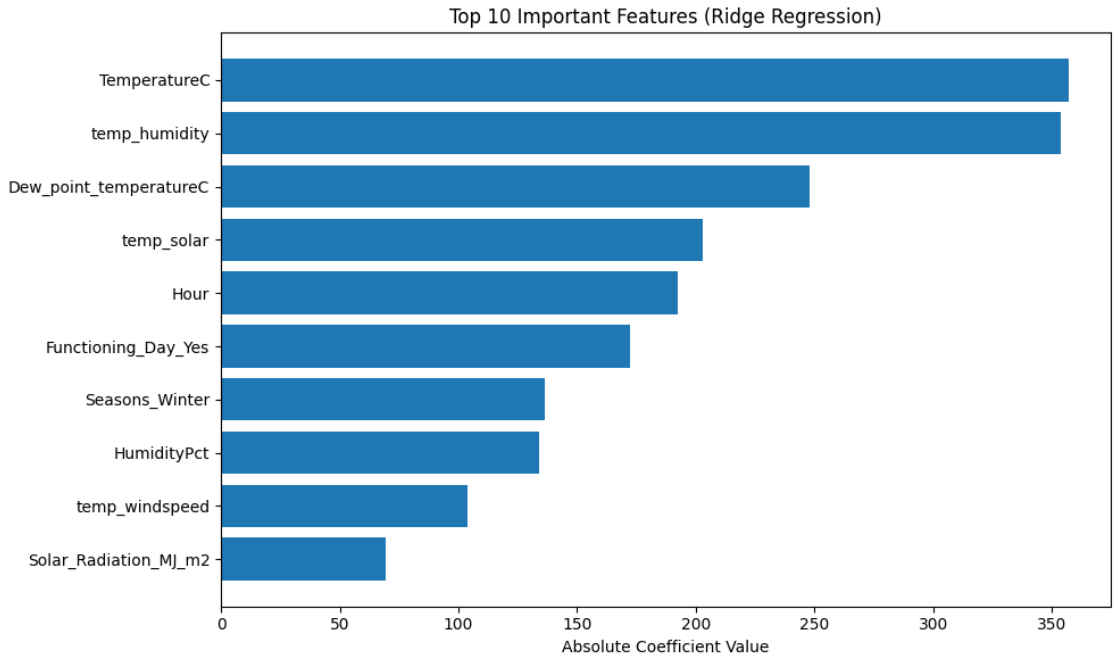
============================================================
TOP 10 MOST IMPORTANT FEATURES
============================================================
                feature  importance
            TemperatureC  357.047633
           temp_humidity  353.952494
Dew_point_temperatureC  247.844873
              temp_solar  203.102079
                    Hour  192.643401
    Functioning_Day_Yes  172.107679
         Seasons_Winter  136.460617
            HumidityPct  134.196785
          temp_windspeed  103.707144
 Solar_Radiation_MJ_m2   69.180521
```

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KEY INSIGHTS AND BUSINESS IMPLICATIONS
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1. MODEL PERFORMANCE:
   - Best Model: Ridge Regression with Polynomial Features
   - Test R² Score: 0.5418
   - This means the model explains 54.2% of the variance in bike rentals

2. KEY PREDICTIVE FACTORS:
   - Temperature is the strongest predictor
   - Hour of day significantly impacts demand (peak hours matter)
   - Weather conditions (humidity, wind, rainfall) affect rentals

3. BUSINESS RECOMMENDATIONS:
   a) Inventory Management:
      - Increase bike availability during peak temperature ranges
      - Adjust staffing based on hourly demand patterns

   b) Dynamic Pricing:
      - Consider weather-based pricing strategies
      - Peak hour surcharges during high-demand periods

   c) Maintenance Scheduling:
      - Plan maintenance during predicted low-demand periods
```

```
      - Use model predictions for weekly/monthly planning

4. MODEL IMPROVEMENTS:
   - Consider adding day-of-week patterns
   - Include seasonal trends more explicitly
   - Explore tree-based models (Random Forest, XGBoost) for potentially better
performance


 Analysis Complete!
```

[ ]:

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